AnalystLab Africa

Machine Learning Internship Program – Batch B

Week 3 Summary Report

Machine Learning Fundamentals

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Dataset Used: Iris Classification Dataset

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1. What I Learned This Week

1.1 Supervised vs Unsupervised Learning

This week I studied the two main branches of machine learning. In supervised learning, the model is trained on labelled data and learns to map inputs to known outputs. I applied this using the Iris dataset – the model was shown 120 labelled flower samples and learned to predict species from petal and sepal measurements.

In unsupervised learning, no labels are provided; the algorithm discovers hidden structure on its own. I applied K-Means clustering to the same Iris data (stripping labels) and observed that the algorithm naturally separated the data into three clusters that closely matched the true species groupings.

Property	Supervised	Unsupervised
Data	Labelled	Unlabelled
Goal	Predict output / class	Find patterns / groups
Algorithms	KNN, Decision Tree, SVM	K-Means, DBSCAN, PCA
Iris Application	Classify species from features	Cluster flowers without labels

1.2 Train / Test Split

I learned that splitting data into training and test subsets is essential for honest model evaluation. I used an 80/20 split (120 training, 30 test samples) with stratification to maintain class balance. Without a test set, a model could achieve perfect training accuracy by memorising examples yet completely fail on real-world data.

1.3 Overfitting and Underfitting

I demonstrated both phenomena using a Decision Tree with varying depths. A depth of 1–2 underfits: the model is too simple to capture species boundaries. A depth of 10+ overfits: training accuracy is perfect but test accuracy plateaus or drops. The ideal depth (~4) balances both concerns and generalises well.

1.4 Model Evaluation Metrics

I applied a K-Nearest Neighbours classifier (k=5) and evaluated it using four core metrics:

- Accuracy – the proportion of all correct predictions (~100% on this dataset)
- Precision – of all flowers predicted as a given species, what fraction truly were that species
- Recall – of all flowers that truly belong to a species, what fraction did the model correctly identify
- F1-Score – the harmonic mean of precision and recall, useful when class sizes differ

I also interpreted a confusion matrix, which showed that the KNN model achieved near-perfect classification with only occasional confusion between versicolor and virginica – the two most morphologically similar species.

2. Challenges Encountered

2.1 Understanding Overfitting Intuitively

Initially it was difficult to grasp why 100% training accuracy could be bad. Plotting training vs. test accuracy across tree depths made the concept visual and concrete: I could see exactly where the model transitioned from underfitting to good fit to overfitting.

2.2 Interpreting the Confusion Matrix

Reading a multi-class confusion matrix required careful attention to rows (actual) vs. columns (predicted). Once I understood the layout, I could pinpoint that all misclassifications occurred at the versicolor–virginica boundary, which makes biological sense as these two species overlap in feature space.

2.3 Connecting Metrics to Business Context

Precision and recall seemed similar at first. The key insight was understanding the cost of each type of error: in a medical diagnosis context, high recall matters more (we do not want to miss a disease), while in spam detection, high precision matters more (we do not want to block legitimate emails).

3. Key Insights Gained

- Even a simple algorithm like KNN can achieve near-perfect accuracy on a clean, well-structured dataset – model complexity is not always the answer.
- Unsupervised K-Means recovered the three Iris species without ever seeing a label, demonstrating the power of pattern recognition in unlabelled data.
- The train/test split is the single most important safeguard against building a model that only works on paper.
- Overfitting and underfitting are two sides of the bias-variance tradeoff: the goal is always to find the sweet spot that generalises.
- Accuracy alone can be misleading; confusion matrices and per-class precision/recall give a much richer picture of model performance.

4. Summary of Topics Covered

Topic	Key Takeaway
Supervised Learning	Labelled data; learns to map inputs to known outputs
Unsupervised Learning	No labels; discovers hidden clusters or patterns
Train/Test Split	80/20 split ensures honest generalisation evaluation
Overfitting	Model too complex; memorises training data
Underfitting	Model too simple; misses patterns in training data

Accuracy	% of correct predictions – good baseline metric
Confusion Matrix	Per-class breakdown of correct and incorrect predictions
Precision & Recall	Quality vs. coverage of positive predictions

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